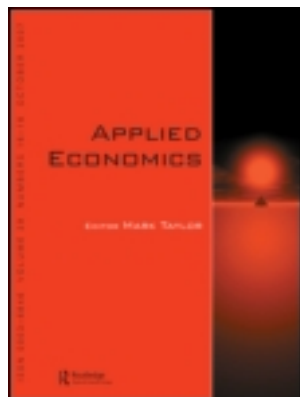




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Bradley T. Ewing^a, Shawn M. Forbes^b & James E. Payne^c

^a Texas Tech University, Lubbock, TX 79409-1014, USA

^b Georgia Southern University, Statesboro, GA 30460-8151, USA

^c College of Business and Technology, Eastern Kentucky University, Richmond, KY 40475-3176, USA

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The effects of macroeconomic shocks on sector-specific returns

BRADLEY T. EWING*, SHAWN M. FORBES† and JAMES E. PAYNE§

Texas Tech University, Lubbock, TX 79409-1014, USA, †Georgia Southern University, Statesboro, GA 30460-8151, USA and §College of Business and Technology, Eastern Kentucky University, Richmond, KY 40475-3176, USA

The reliance on market and sector-specific indexes to evaluate managed portfolios and the popularity of index investing has increased the importance of understanding what leads to market movements, how long they may last, and how different sectors respond to macroeconomic shocks. This research is concerned with how shocks to macroeconomic variables affect five major S&P sector-specific stock market indexes. The paper employs the newly developed econometric technique of *generalized* impulse response analysis. The results identify the various responses of the sectors to unanticipated changes in some key macroeconomic variables.

Asset prices are commonly believed to react sensitively to economic news. Daily experience seems to support the view that individual asset prices are influenced by a wide variety of unanticipated events and that some events have a more pervasive effect on asset prices than do others. (Chen *et al.* 1986, p. 386)

I. INTRODUCTION

The above quote suggests that macroeconomic variables affect stock returns, but that these effects may vary in terms of magnitude and persistence. A simple extension to this is that unanticipated changes in certain macroeconomic variables influence assets comprising a market index in a predictable fashion, but that the unanticipated event may have a more pervasive effect on the returns of some stock indexes than it does on others. In today's investment world, financial market participants are interested not only in how individual stocks perform, but also in how various market indexes perform. Individual investors, mutual fund managers, and numerous institutional investors use market indexes as benchmarks to track the

performance of publicly traded stocks and actively managed portfolios. Many indexes are used as the basis for a variety of financial instruments such as options, warrants, derivatives, insurance products and index annuities, as well as index mutual funds and institutional indexed accounts.¹ The popularity of index funds as an integral part of the portfolio allocation decision together with the reliance on overall and sector-specific indexes to evaluate managed portfolios has increased the importance of tracking various indexes in the financial press.²

Following in the footsteps of Chen *et al.* (1986) and others, this paper seeks not only to identify, but to examine, how innovations in certain key macroeconomic variables are transmitted to several stock market indexes. In particular, the magnitude and persistence of the

* Corresponding author. E-mail: Bradley.Ewing@ttu.edu

¹ Many mutual fund companies offer index funds and other index-linked products. For example, more than 50 companies now issue S&P 500 index-linked annuities. <http://www.spglobal.com/index.html>.

² Speaking on *Louis Rukeyser's 1999 Money Guide* show on predicting trends, noted Wall Street expert Don Phillips suggested that every investor ought to own an S&P stock index fund.

responses of five major S&P sector-specific stock market indexes to unexpected changes (i.e., shocks) to real output, the stance of monetary policy, and risk premia are looked at.³ It has been found that these macroeconomic variables are associated with stock market behaviour. In fact, Ewing (2001a) determined that shocks to these three factors explained a significant portion of the future changes in the S&P composite index. Current stock returns might reflect expected future changes in output as well as some degree of contemporaneous correlation (Chen *et al.*, 1986; Chen, 1991). The relationship between changes in stock prices and measures of monetary policy might arise if money has real effects (Thorbecke, 1997) or if monetary policy represents a significant source of business cycles (Patelis, 1997). Changes in risk premia might be linked to stock returns as a measure of risk aversion, level of market risk, and possibly through a 'leverage effect' as well (Chen *et al.*, 1986). However, sector-specific responses to macroeconomic shocks may not be identical due to underlying differences in market and industrial characteristics. Thus, this study differs from that of Ewing (2001a) by focusing on returns in the capital goods, financials, industrials, transportation, and utilities sectors instead of the overall market. The present study does not impose the a priori restriction that all sectors behave similarly, as is implicitly assumed in much of the existing literature that examines composite market returns. Instead, the method used allows the returns in various sectors to move independently of other sectors in response to macroeconomic shocks, even though sector returns may be contemporaneously correlated.

To conduct the analyses, vector autoregressions are estimated and *generalized* impulse response functions computed (Koop *et al.*, 1996; Pesaran and Shin, 1998). These response functions allow the effects of unanticipated changes in the macroeconomic variables to be compared and contrasted, which can be interpreted as unexpected 'economic news,' on the indexes. Clearly, the potential exists for many financial market participants to be affected by movements in market indexes. A better understanding of what leads to market movements, how long they may last, and how different sectors respond to macroeconomic shocks would be helpful.

II. A BRIEF REVIEW OF PAST RESEARCH

Many researchers have determined that macroeconomic variables play an important role in determining stock prices. In one of the most influential papers in this area, Chen *et al.* (1986) examine the effects of macroeconomic

variables on stock market returns and find that many macroeconomic sources of risk, including output and risk premia, are significantly priced. Stokes and Neuburger (1998) consider the effects that macroeconomic activity has on an aggregate measure of stock returns and conclude that these variables may play a significant role in the stock market. Examining the relationship between monetary policy and the stock market, Patelis (1997) and Ewing (2001b) found that monetary policy variables were significant predictors of stock market returns. Thorbecke (1997) found that innovations to the fed funds rate influenced stock prices. Based on the theory that stock prices reflect the present value of future net cash flow, he concludes that monetary policy has real effects by increasing future cash flows or by decreasing discount factors used in the discounted cash flow model.

Others have found macroeconomic variables are important in regards to stock markets. In one of the best known papers, Friedman (1988) found a significant relationship between M2 money and stock prices. In two related papers, Lee (1995, 1998) used aggregate price-dividend data and found that price-dividend spreads are primarily accounted for by temporary shocks to dividends. He also concluded that substantial variation in stock prices is due to temporary shocks. Examining annual stock index, aggregate dividend, and earnings data in a VAR framework, Lee found that deviations of stock prices from these fundamentals is quite substantial. Using a VAR framework that included orthogonalized impulse response analysis, Ewing *et al.* (1998) documented the relationship among interest rates comprising the spread between lending and deposit rates and the S&P financial stock index. Also using the VAR methodology, Lee (1992) examined the linkages among stock market returns, interest rates, real activity, and inflation and concluded that stock returns may Granger-cause real activity. The relationship between stock returns and macroeconomic variables was also considered by Hess and Lee (1999) in the context of aggregate supply and demand shocks. Using VAR techniques they determined that unexpected inflation can have either negative or positive effects on stock returns depending on the source of inflation. Related work by He and Ng (1994) incorporated size and book-to-market values of equity as measures of macro risk when examining the relationship between market fundamentals, economics forces, and the stocks. As the above literature attests to, stock market indexes are linked to macroeconomic variables. However, unlike previous studies, shocks to various macroeconomic factors are allowed to affect returns by sector differently. Thus, this research complements the existing literature that has primarily focused on overall market returns. The paper fills a

³ The Standard and Poor's indexes are particularly interesting to examine as, according to Mennis (1999), financial professionals use the S&P indexes more than any others to follow the movement of industry groups.

void in this line of research and provides insight into the responses of five major sector-specific stock market indexes to macroeconomic shocks.

III. DATA

The response of S&P sector-specific stock market indexes to shocks to real output, monetary policy, and market risk premium are examined, over the post-crash period 1988:1–1997:7.⁴ Following Estrella and Mishkin (1997), the coincident index is used as a proxy for real output. Based on the work of Bernanke and Blinder (1992), Patelis (1997), and Thorbecke (1997) the federal funds rate is used as a measure of the stance of monetary policy. As a measure of risk premia the spread between low grade corporate bond (Baa) and 10-year government Treasury bond rates is used (see Chen *et al.*, 1986; Bernanke and Blinder, 1992). Additionally, the recent study of the market composite index by Ewing (2001a) included these three macroeconomic factors. All variables except interest rates are entered in natural logarithms. Thus, the monthly data include the index of coincident indicators (*LNCOINC*), the federal funds rate (*FYFF*), and the interest rate spread (*RISK*) as well as five sector-specific S&P stock market price indexes: capital goods (*LNCAP*), financials (*LNFI*), industrials (*LNIN*), transportation (*LNTR*), and utilities (*LNUT*).⁵ All data are from DRI/Citibase.

IV. THE VAR ANALYSIS AND EXAMINATION OF DYNAMIC RESPONSES

Examination of the dynamic relationships among variables in vector autoregressive (VAR) models can be conducted using innovation accounting methods such as impulse response functions. The conventional method of Sims (1980) has been criticized because the impulse responses may appear markedly different depending on the ordering of the variables in the VAR (Lutkenpohl, 1991). This paper employs the recently developed ‘generalized’ impulse response function (Pesaran and Shin, 1998; Koop *et al.*,

1996) that is not sensitive to the ordering of the variables in the VAR.

Consider the following moving average representation of the VAR (m) model:

$$\mathbf{X}_t = \Psi(L)\mathbf{v}_t \quad (1)$$

where $E(\mathbf{v}_t\mathbf{v}_t') = \Sigma_v$ such that shocks are contemporaneously correlated, and L is a polynomial in the lag operator. The generalized impulse response function of X_i (e.g. a sector return) to a unit (one standard deviation) shock in X_j (e.g., a macroeconomic factor) is given by:

$$\Psi_{ij,N} = (\sigma_{ii})^{-1/2}(\mathbf{e}_j'\Sigma_v\mathbf{e}_i) \quad (2)$$

where σ_{ii} is the i th diagonal element of Σ_v , $\mathbf{e}_i$ is a selection vector with the i th element equal to one and all other elements equal to zero, and N is the horizon.⁶

Two key elements of the generalized impulse response function stand out. First, the generalized responses are invariant to any re-ordering of the variables in the VAR.⁷ Thus, the generalized impulse response function provides more robust results than the orthogonalized method. Second, because no orthogonality assumption is imposed, the generalized impulse response function allows the researcher to examine the initial impact response of each variable to shocks to any of the other variables. The ability of the generalized impulse response function to capture these immediate responses of endogenous variables to shocks is clearly advantageous, especially in studies of equity markets when information is very often quickly assimilated.

In preliminary analysis it was determined that all variables except the interest rate spread (*RISK*) require first-differencing to attain stationarity; however, evidence of cointegration among the variables was not found.⁸ Thus, the first-differences of *LNCOINC*, *FYFF*, *LNCAP*, *LNFI*, *LNIN*, *LNTR*, and *LNUT* are used and the level of *RISK* in the VAR estimations used to generate the generalized impulse responses in the empirical analysis. The first difference operator is denoted as ‘D’. Table 1 presents descriptive statistics for the variables. Note that annualized returns based on the monthly changes in the sector-specific indexes range from a low of 6.68% in utilities to a high of 16.30% for financials. Similarly, there is wide array of variation in the returns by sector. The smallest standard

⁴ Focus is on the post-1987 crash period because a number of studies have documented differences in stock market behaviour when comparing pre- and post-crash periods (Lee and Kim, 1993–1994; Choudhry, 1996; Meric and Meric, 1997; Aggarwal *et al.*, 1999).

⁵ The S&P indexes are market weighted with base period 1941–1943 set to 10.0. Complete descriptions of the indexes can be found on the S&P website.

⁶ Recall that the traditional orthogonalized impulse response employs a Cholesky decomposition of the positive definite covariance matrix. See Patelis (1997) for details. The generalized version does not impose this restriction.

⁷ Pesaran and Shin (1998) note that ‘generalized impulse responses are unique and fully take account of the historical patterns of correlations observed amongst the different shocks’ (p. 20). The generalized and orthogonalized impulse responses coincide only when the covariance matrix is diagonal.

⁸ ADF and Phillips–Perron unit root test results as well as Johansen–Juselius cointegration test results are available on request from the authors.

Table 1. Descriptive statistics (adjusted sample period is February 1988–July 1997)

	Mean	Median	Standard deviation
<i>DLNCAP</i>	12.33	10.76	42.61
<i>DLNFI</i>	16.30	16.02	53.39
<i>DLNIN</i>	13.99	13.93	32.66
<i>DLNTR</i>	13.02	16.48	47.55
<i>DLNUT</i>	6.68	13.28	32.26
<i>DLNCOINC</i>	2.11	2.43	3.19
<i>DFYFF</i>	−0.01	−0.01	0.21
<i>RISK</i>	0.83	0.78	0.20

Note: *DLNCAP*=annualized monthly return on the capital goods index, *DLNFI*=annualized monthly return on the financials index, *DLNIN*=annualized monthly return on the industrials index, *DLNTR*=annualized monthly return on the transportation index, *DLNUT*=annualized monthly return on the utilities index, *RISK*=spread between Baa and long-term government bond rates, *DFYFF*=change in the fed funds rate, and *DLNCOINC*=annualized monthly growth rate in the coincident index.

deviations appear in industrials and utilities while the financials exhibits the largest standard deviation. In statistical tests it was not possible to reject the null hypotheses of equal means and equal variances. Plots of the return series, though not included here, also revealed that returns by sector, though similar in some respects, are clearly not identical.⁹ Based on these cursory findings, further analysis is warranted.

Three separate vector autoregressions are estimated. Each VAR has six equations and contains six variables, *DLNCAP*, *DLNFI*, *DLNIN*, *DLNTR*, *DLNUT*, and either *DLNCOINC*, *DFYFF*, or *RISK*, in addition to a constant term.¹⁰ These will be referred to as VAR1, VAR2, and VAR3, respectively. The optimal order of each VAR was determined to be one based on Akaike's information criterion, Schwartz Bayesian criterion, and likelihood ratio tests. As discussed above, one of the major concerns and potential drawbacks of using orthogonalized impulse responses is that results might be sensitive to the ordering of the variables. Indeed, if the shocks to the respective equations in a VAR are contemporaneously correlated, then the orthogonalized and generalized impulse responses may be quite different. Moreover, a re-ordering of the variables may lead to a number of markedly different conclusions based on the orthogonalized responses. If, on the

other hand, shocks are not contemporaneously correlated then the two types of responses would probably not be that different and the orthogonalized responses would probably not be very sensitive to a re-ordering of the variables. Thus, before proceeding to an examination of the dynamic responses of the respective stock indexes to unanticipated changes in macroeconomic variables, a test was performed to determine if shocks in the six individual equations that make up each of the three VARs were contemporaneously correlated. The null hypothesis is that the off-diagonal elements in the covariance matrix equal zero and is tested against the alternative that none of the off-diagonal elements is equal to zero. The log-likelihood ratio test statistic is computed as $LR = 2(LL_u - LL_r)$ where LL_u and LL_r are the maximized values for the log-likelihood functions for the unrestricted and restricted models, respectively.¹¹ The LR statistics are distributed χ^2 with 6 degrees of freedom and were 481 for VAR1, 464 for VAR2, and 494 for VAR3. Each of the LR statistics was significant at less than the 1% level. Thus, the null is rejected for each VAR and we proceed to examine generalized impulse responses.

V. DISCUSSION OF RESULTS

Figures 1–3 present the results of the generalized impulse response functions for the three VARs. The graphs trace out the response of changes in each of the S&P indexes for a one standard error (positive) shock to changes in the log of the coincident index (*DLNCOINC*), changes in the fed funds rate (*DFYFF*), and the risk premium (*RISK*), respectively. Responses are plotted out to the fifth month.

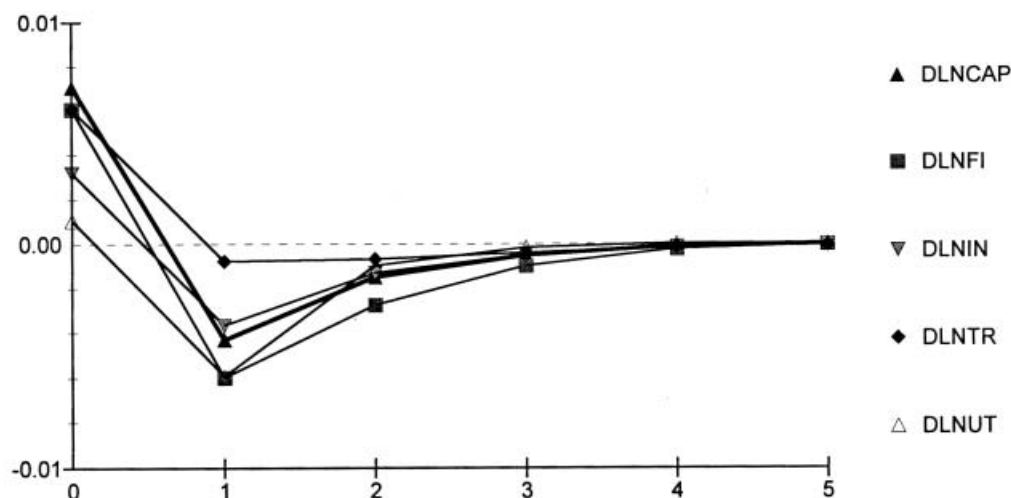
As can be seen in Fig. 1 for each of the S&P stock index returns, there is a positive and significant initial impact (i.e., at $N = 0$) for each of the returns in response to an unanticipated increase in real output growth.¹² However, immediately after initial impact (e.g., at $N + \alpha > 0$, where α is an arbitrarily small positive number) there is a decrease in the responses, and by the end of the first period the response is negative and statistically significant for each of the index returns. What this suggests is that for these S&P sectors a positive shock to this macroeconomic variable is quickly translated into higher returns but the effect quickly wears off leading to an overshooting of the baseline levels before the effect dies off completely. Financial

⁹ The results from the tests of equality of means and equality of variances, as well as the return plots are available upon request.

¹⁰ One VAR was also estimated that included all eight variables and the impulse responses were virtually indistinguishable from those reported in the paper.

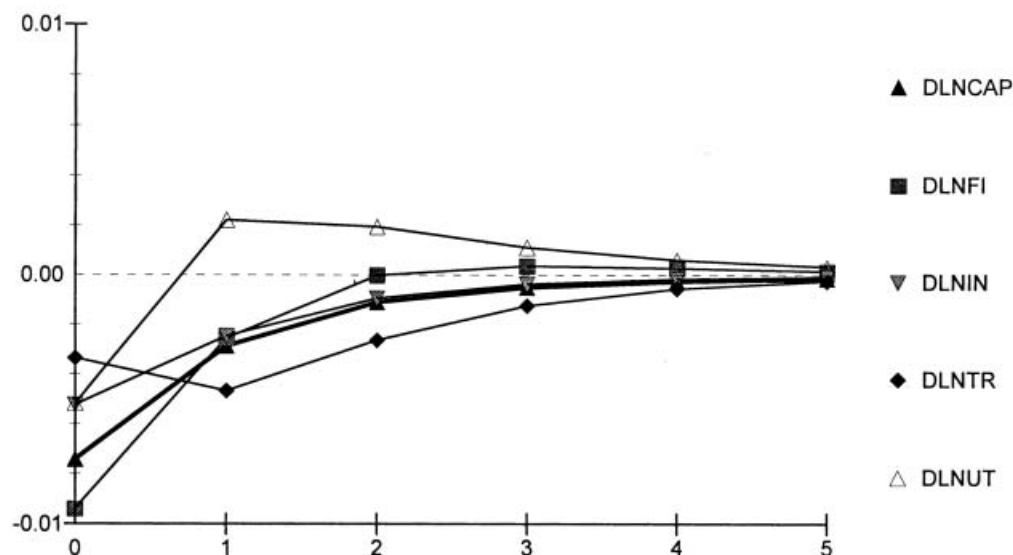
¹¹ LL_u is the system log-likelihood from the VAR and LL_r is computed as the sum of the log-likelihood values from the individual equations in the VAR.

¹² Significance is determined by the use of confidence intervals representing two standard deviations. In order to keep the graphs legible, these bands are not illustrated. This convention is used throughout the discussion of results. See Runkle (1987) for a discussion on confidence intervals.



All responses are significant at $N = 0$; all responses are significant at $N = 1$ except *DLNTR*

Fig. 1. Generalized impulse responses to a shock to output growth (DLNCOINC)



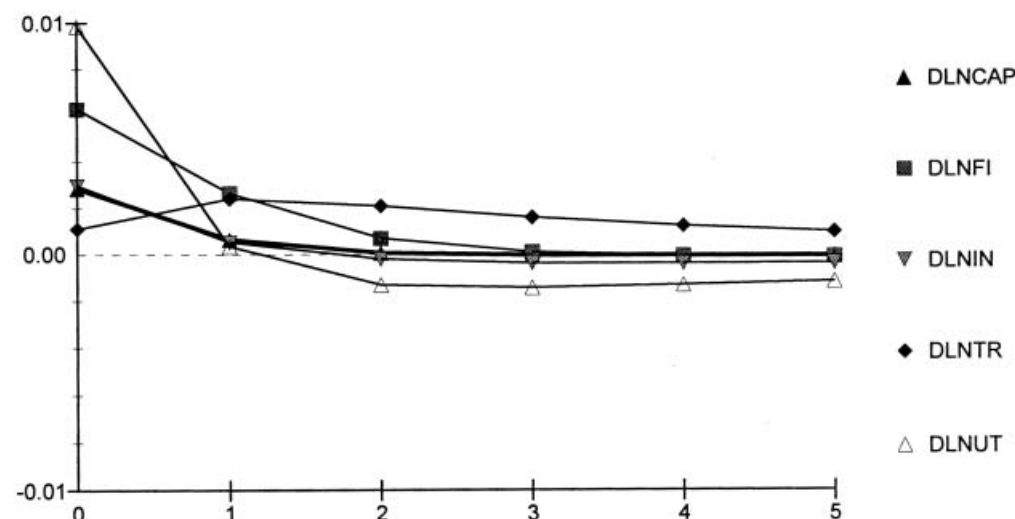
All responses are significant at $N = 0, 1$; all responses are significant at $N = 2$ except *DLNFI*

Fig. 2. Generalized impulse responses to a shock to monetary policy (DFYFF)

market participants find this unexpected economic growth to be attractive and it is rewarded in each sector of the market. While the general behaviour is the same for each index, the magnitudes of initial impact, the amount of overshooting that occurs, and the persistence of the output shocks all vary by index. For instance, the greatest initial impact effect occurs in capital goods (0.0070), followed by financials (0.0061), transportation (0.0060), industrials (0.0032), and utilities (0.0010). The amount of overshooting that occurs far exceeds the absolute value of the initial impact for utilities, slightly exceeds it for industrials, and is far less dramatic for capital goods and is barely discernible for transportation. Note that the greater the overshooting,

the more volatile the index is in response to the shock. The output effect wears off the quickest in utilities and capital goods, and lasts the longest in industrials.

Figure 2 shows the generalized responses computed from VAR2. The initial impact effect of an unanticipated monetary tightening is negative and statistically significant for each of the indexes. The effects, from smallest to most severe, occur in transportation (−0.0029), utilities (−0.0050), industrials (−0.0049), capital goods (−0.0071), and financials (−0.0092). The initial impact effect is followed by an overshooting only for the case of utilities. Each response is statistically significant for $N = 0, 1$ and remain significantly different from zero for $N = 2$,



All responses are significant at $N = 0$; responses for *DLNFI* and *DLNTR* are significant at $N = 1$ while responses for *DLNCAP*, *DLNIN*, and *DLNUT* are not significant at $N = 1$

Fig. 3. Generalized impulse responses to a shock to risk premia (RISK)

except for the case of financials. No overshooting effect is detected in the capital goods, industrials, and transportation indexes. The fed funds effect essentially dies out by the second month after the shock for capital goods, and industrials, but lingers on (marginally) for another period for transportation and utilities.

The risk premium effect is shown in Fig. 3 which presents the generalized responses based on the estimation of VAR3. A sudden positive shock to *RISK*, the spread between Baa and long term government bond rates, is translated into positive and statistically significant initial impacts for each of the S&P indexes. By far and away, the largest effect is felt in utilities (0.0099), followed by financials (0.0063), capital goods and industrials (0.0028 each), and transportation (0.0011). An overshooting only occurs in utilities but not until after the first period before dying off. The responses are also significant at $N = 1$ for financial and transportation. The risk premium shock wears off in about two months for financials. The effect is actually greater at the end of one period than at initial impact for transportation and persists slightly longer.

Generally speaking, shocks to the three key macroeconomic variables investigated have initial impacts consistent with what could be explained by simple financial theory and are in line with the findings of others. In this respect, the research has broken down these effects by various economic sectors using five S&P market indexes. The initial impacts to various shocks are not identical across these sectors. It is also found that macroeconomic shocks lead to greater volatility in some indexes than in others. Moreover, the effects of macroeconomic shocks die out at different rates, lasting far longer or wearing out much quicker, depending on the index examined.

In summary, unanticipated changes in economic growth lead to similar patterns in the S&P sector-specific indexes, with the effects differing mainly in terms of the magnitude of initial impact and the amount of overshooting and, thus, volatility that occurs following the shock. Utilities barely register an initial impact but the overshooting is quite noticeable, while the opposite is true for transportation. Unexpected real output growth places pressure on the productive capacity of the economy. In the standard IS-LM model, in conjunction with a price adjustment mechanism such as the expectations-augmented Phillips curve, an increase in aggregate demand growth raises output, thus placing pressure on prices to rise. Consequently, positive initial impacts are observed as investors reward the 'good news' associated with growth shocks. However, the expectation of future higher prices and the potential for declining output then places downward pressure on stock returns. The effects are felt most in the industries that exhibit the least control over pricing or where price competition is greatest.

A monetary policy shock has its largest effect on financials and capital goods, and seems to lead to a short but dramatic cyclical volatility in utilities. Not surprisingly, financial firms are found to be extremely sensitive to changes in monetary policy due, of course, to the subsequent changes in lending and deposit rates that follow changes in the fed funds rate. The findings are consistent with capital goods firms being among the industries that are most responsive to changes in interest rates due, in part, to the higher debt obligations of these types of firms. The highly regulated nature of the utilities industry may restrict these firms from responding quickly to unanticipated changes in monetary policy. These constraints may lead

to lags in implementing strategies and making changes and, therefore, result in the overshooting and overreaction in their stock prices.

Risk premium changes appear to effect utilities and transportation much differently than they do financials, capital goods, and industrials. The findings are consistent with restrictive environments making it more difficult for firms to manage risk. In the case of utilities, the restrictions are highly regulatory. Transportation firms, which are also subject to heavy regulation, face constraints from energy markets and are highly susceptible to changes in petroleum prices.

CONCLUDING REMARKS

This paper has examined and documented the responses of five sector-specific S&P stock price indexes to shocks to three key macroeconomic variables using the newly developed technique of generalized impulse response analysis. The technique allows for robust results and enables comparison of both the severity and extent of macroeconomic shocks to these indexes.

The results add to the literature on the linkages between stock markets and the macroeconomy, but do not impose the a priori restriction that all sectors behave similarly, as is implicitly assumed in much of the existing literature that examines composite market returns. Instead, the method allows the returns in various sectors to move independently of other sectors in response to macroeconomic shocks, even though sector returns may be contemporaneously correlated. The findings are especially interesting given the importance and popularity of sector-based stock market indexes and investment strategies in today's financial markets.

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